

Data Visualization with Esquisse

Esquisse Package

```
# install.packages("esquisse")  
library(esquisse)
```

Esquisse Package

The [esquisse package](#) is helpful for getting used to creating plots in R.

It is an interactive tool to help you in RStudio.

It's super **nifty**!



First, get some data..

We can use the CO heat-related ER visits dataset. This dataset contains information about the number and rate of visits for heat-related illness to Emergency rooms in Colorado from 2011-2022, adjusted for age.

```
er <-  
  read_csv("https://jhudatascience.org/intro_to_r/data/CO_ER_heat_visits.csv")
```

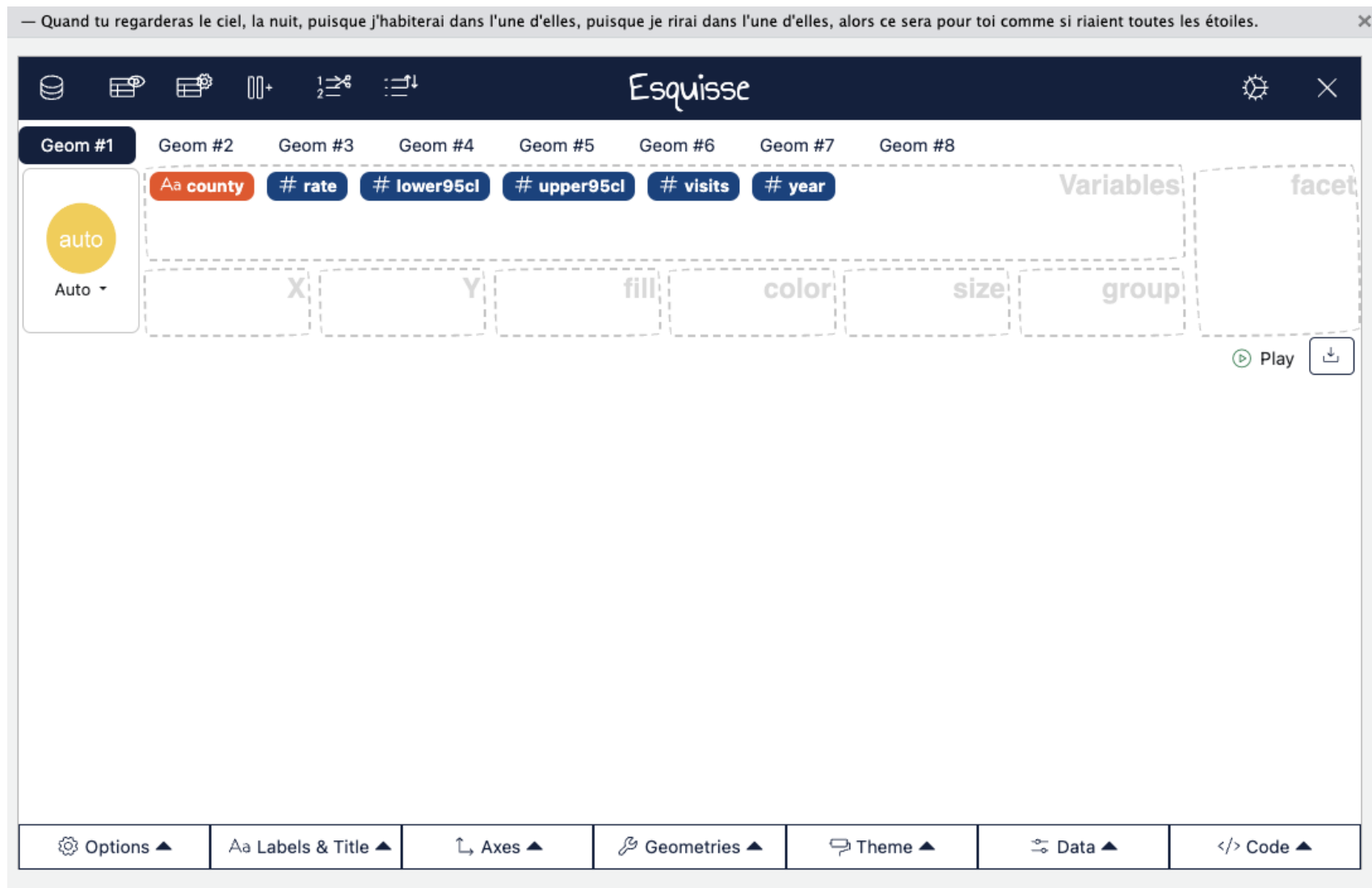
```
head(er)
```

```
## # A tibble: 6 × 6  
##   county rate lower95cl upper95cl visits year  
##   <chr> <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 Adams 6.73 NA 9.24 29 2011  
## 2 Adams 4.84 2.85 NA 23 2012  
## 3 Adams 6.84 4.36 9.31 31 2013  
## 4 Adams 3.08 1.71 4.85 15 2014  
## 5 Adams 3.36 1.89 5.23 16 2015  
## 6 Adams 8.85 6.12 11.6 42 2016
```

Starting a plot

Using the `esquisser()` function you can start creating a plot for a `data.frame` or `tibble`. That's it!

`esquisser(er)`

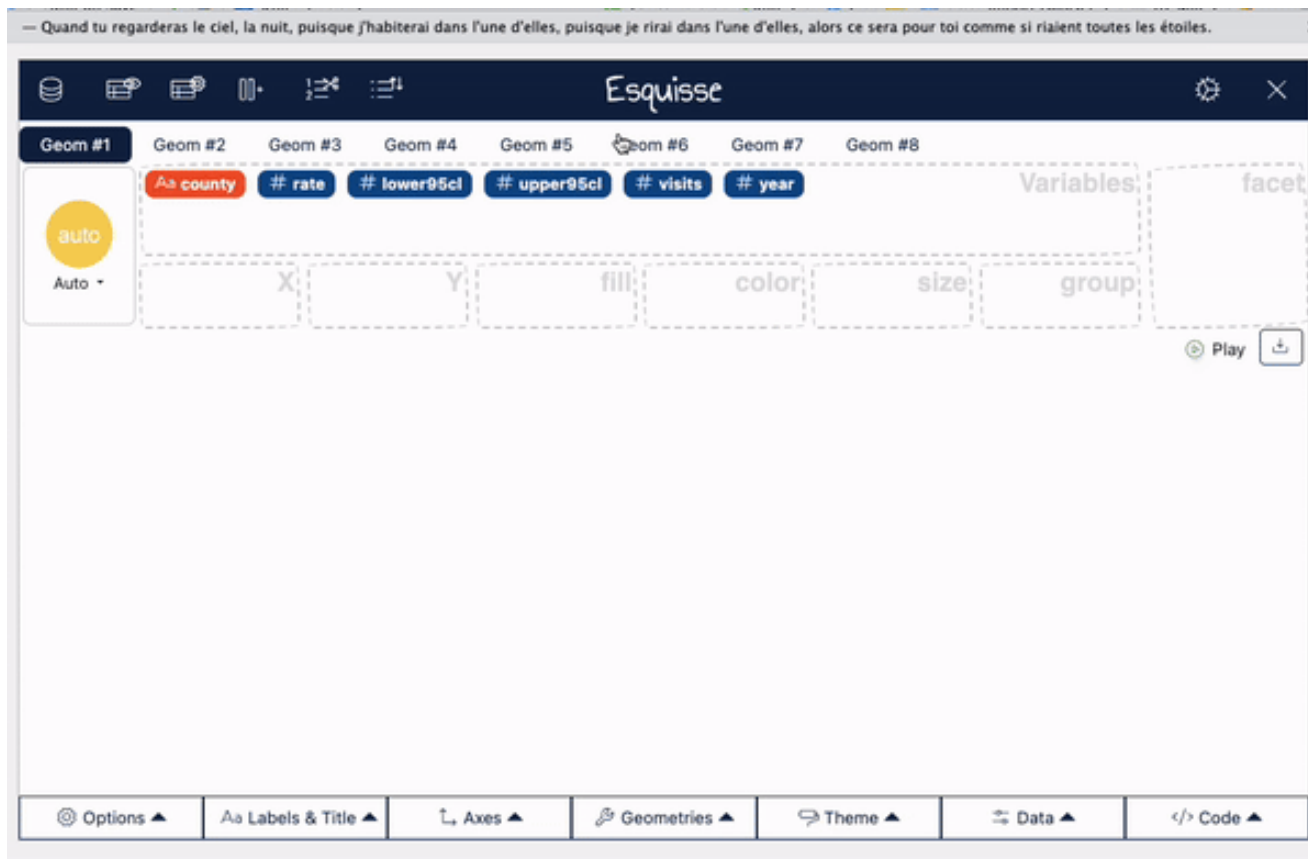


Show the plot in the browser

```
esquisse::esquisser(er, viewer = "browser")
```

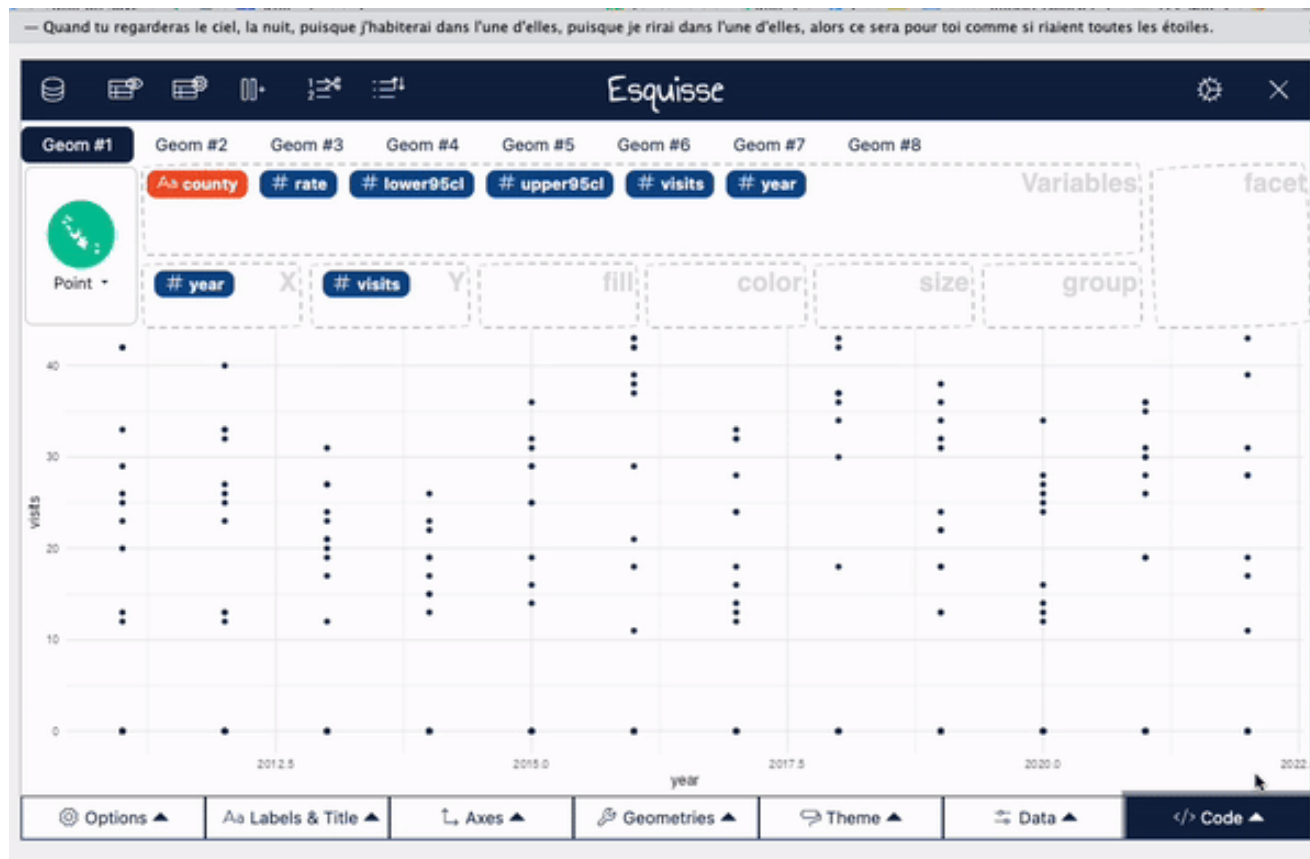
Select Variables

To select variables you can drag and drop variables to the respective axis that you would like the variable to be plotted on.



Find code

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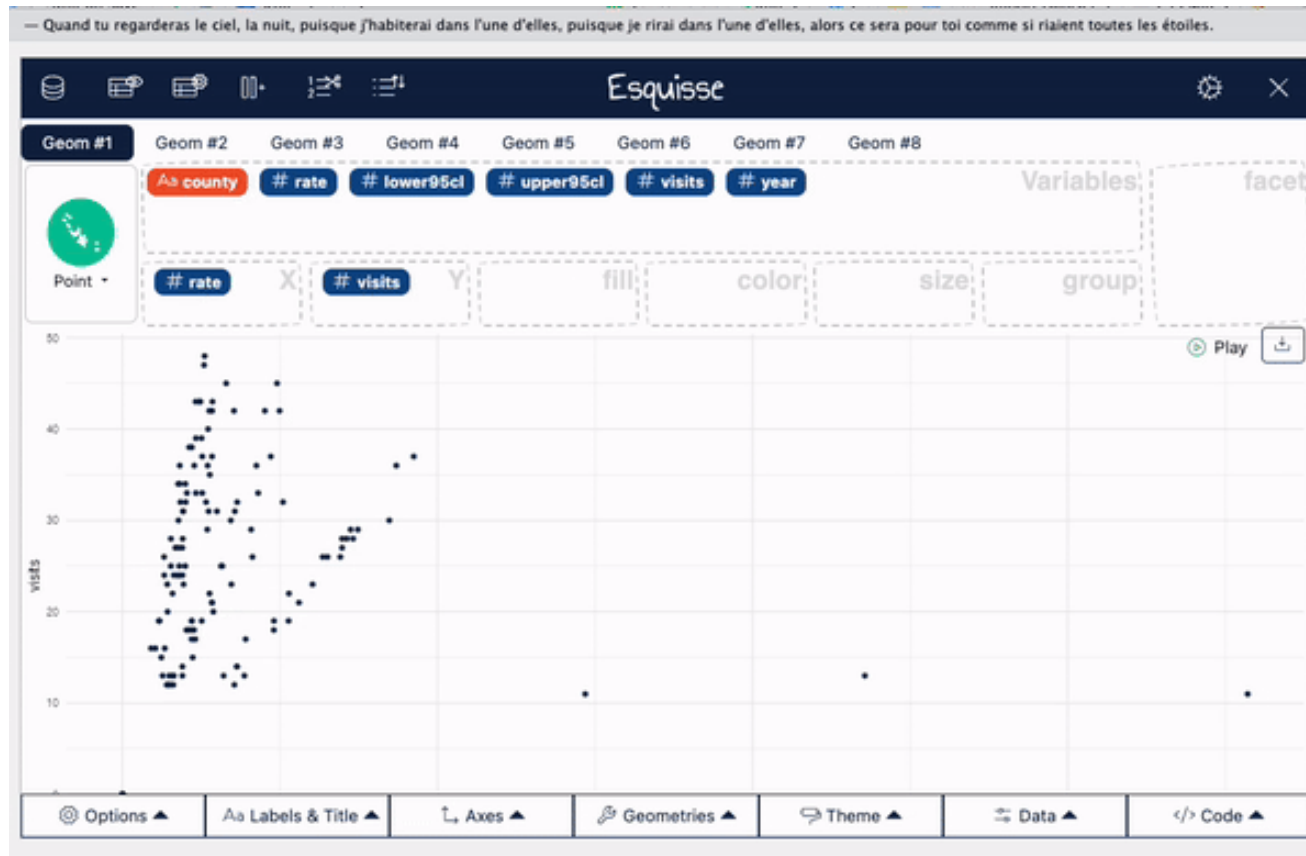
Change plot type

esquisse automatically assumes a plot type, but you might want to change this.



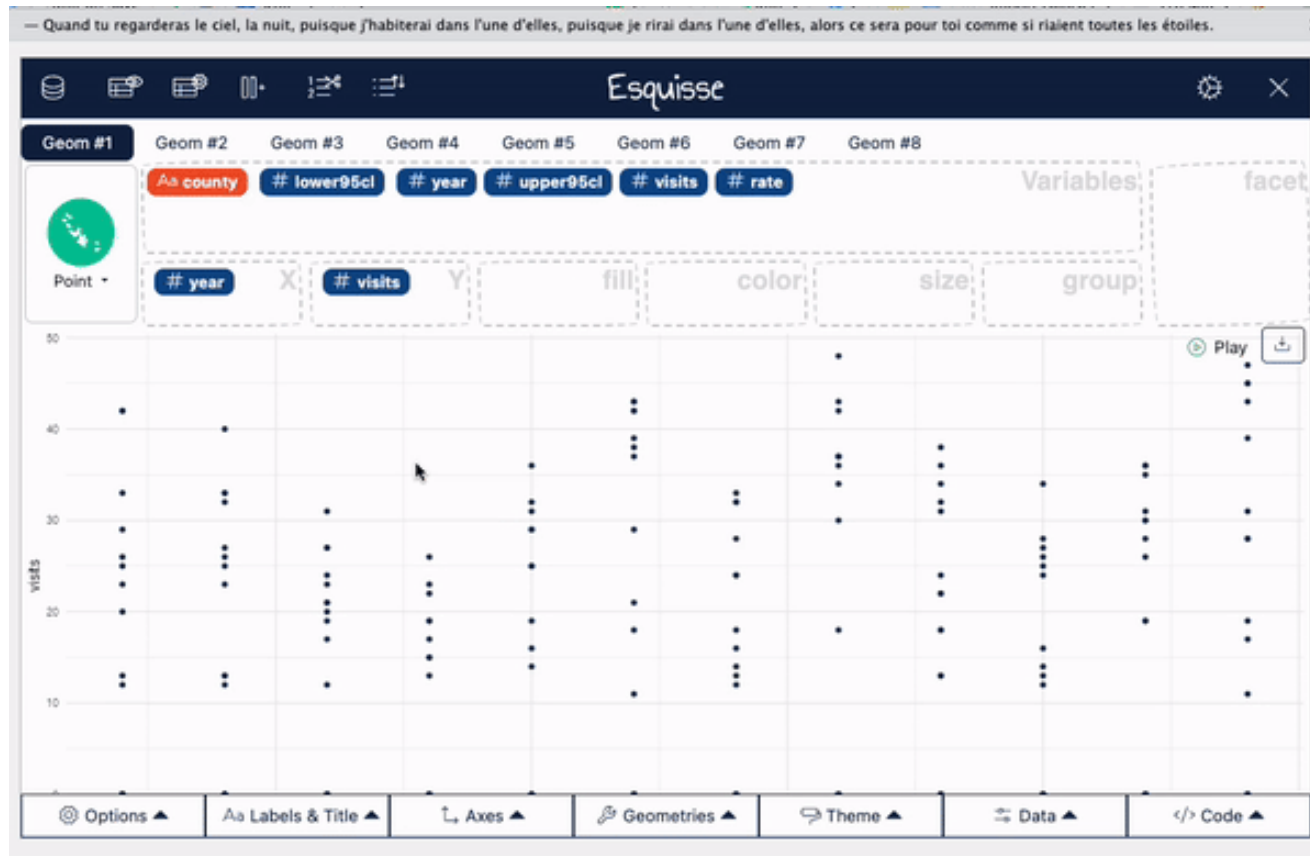
Add Facets

Facets create multiple plots based on the different values of a variable.



Add size

Sometimes it is useful to change the way points are plotted so that size represents a variable. This can especially be helpful if you need your plot to be black and white.



Add color

For plots with points use the color region to change coloring according to a variable. (use “fill” for bar plots)



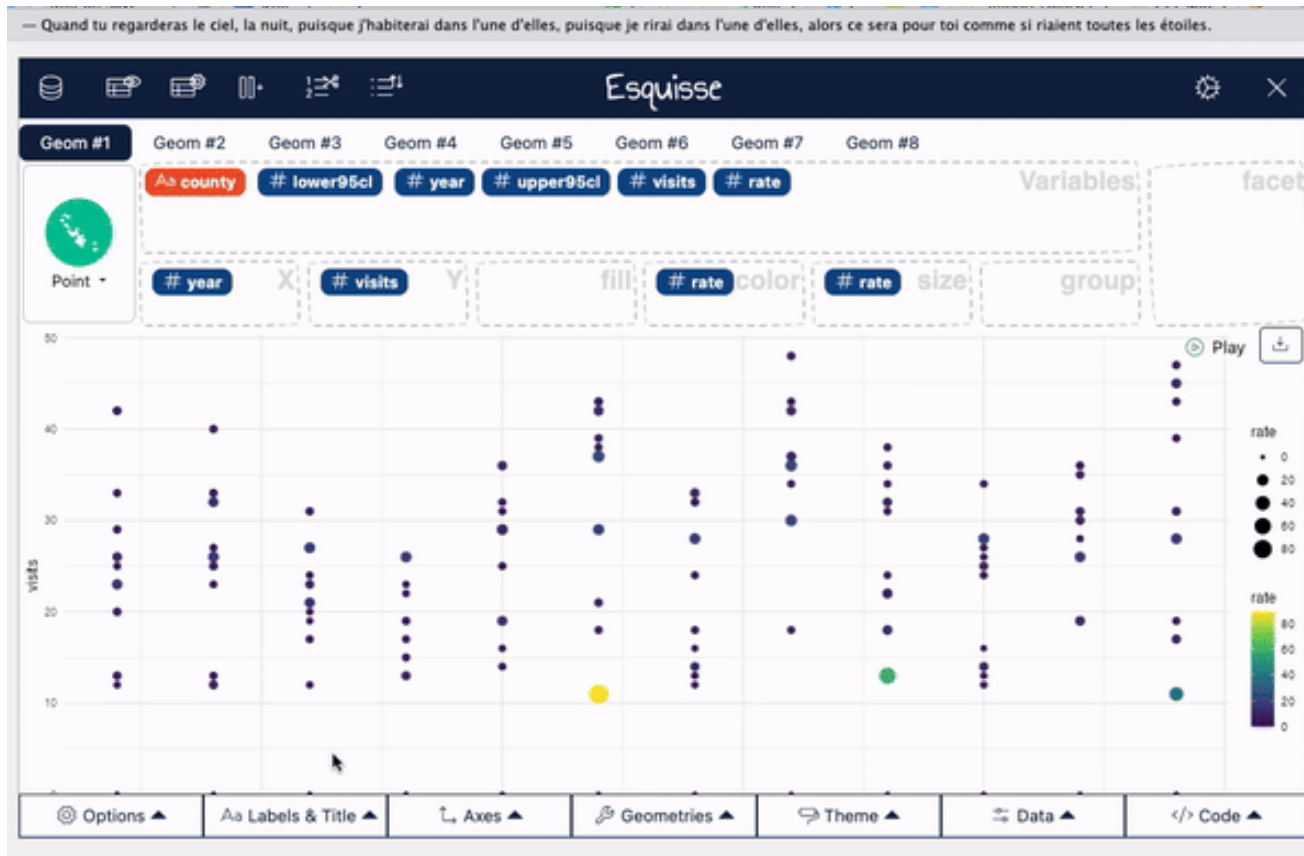
Appearance

You can change the overall appearance with “Geometries” and “Theme”.



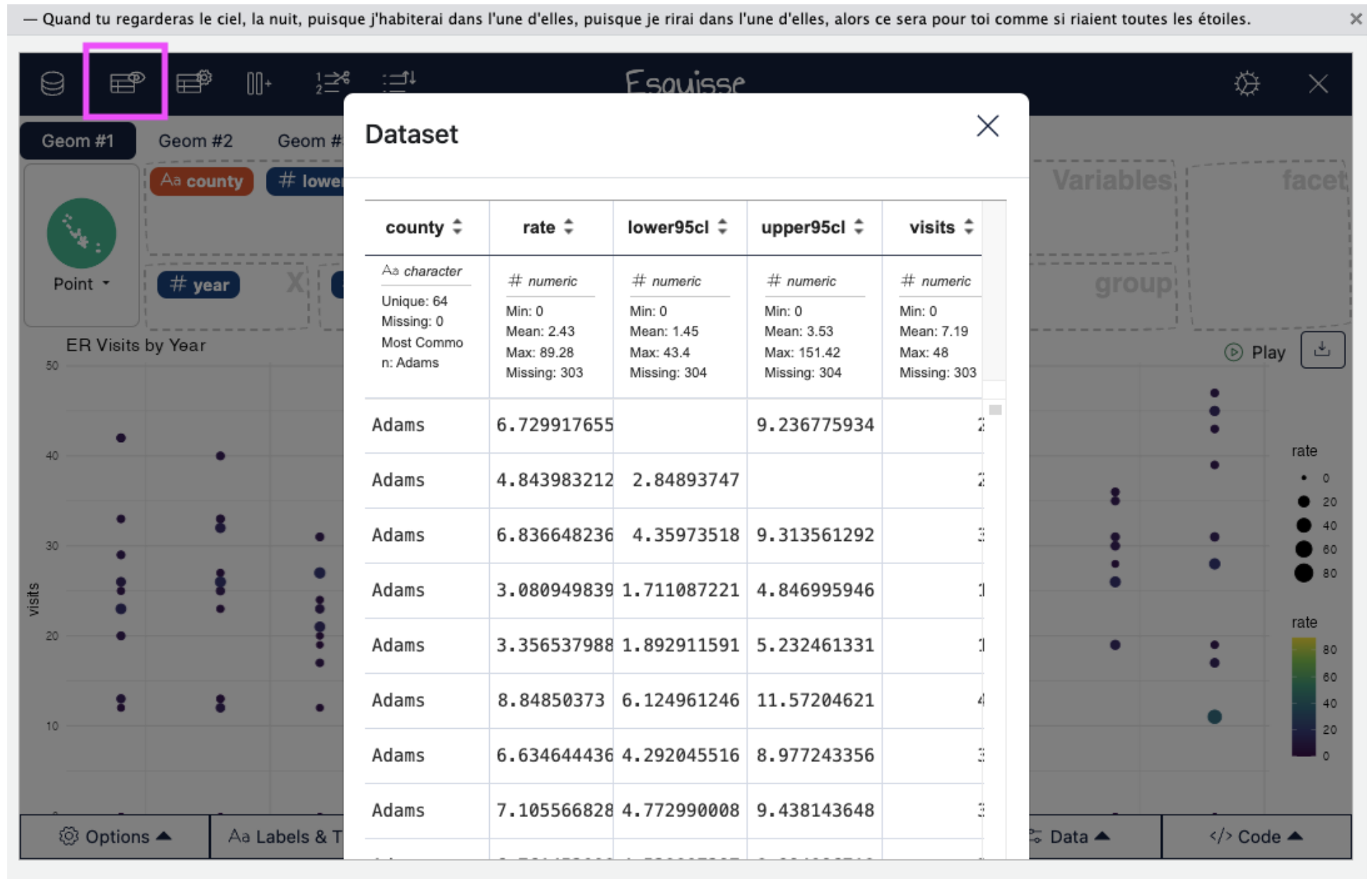
Change titles

To change titles on your plot, use the “Labels & Titles” tab.



View data

You can also easily view data

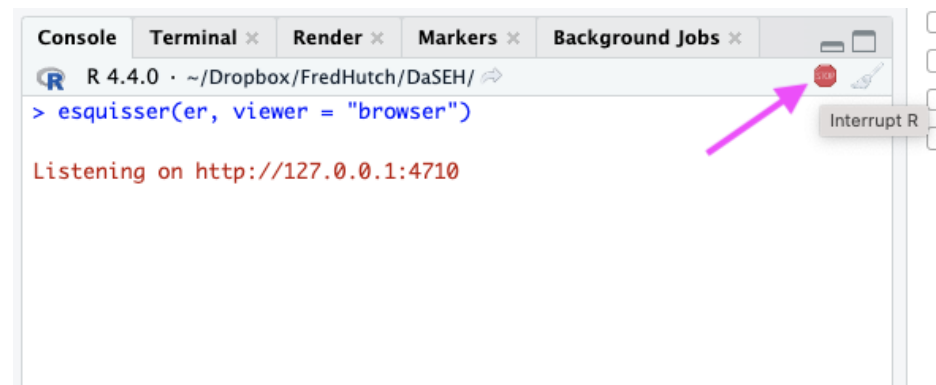


Interrupting Esquisse

You'll need to "interrupt" Esquisse to launch it with a new dataset.

Use the stop button or press ctrl+c to stop the Esquisse app.

If you don't see the stop button, you need to resize your window.



Wide & Long Data ?

Let's look at why we might want long data using Esquisse.

```
library(tidyverse)
er <- read_csv(file =
  "https://jhudatascience.org/intro_to_r/data/CO_ER_heat_visits.csv")
long_er <- er %>%
  filter(county == c("Denver", "Boulder")) %>%
  select(c("county", "year", "visits"))
glimpse(long_er)

## Rows: 12
## Columns: 3
## $ county <chr> "Boulder", "Boulder", "Boulder", "Boulder", "Boulder", "Boulder..."
## $ year   <dbl> 2012, 2014, 2016, 2018, 2020, 2022, 2011, 2013, 2015, 2017, 2019, 2021...
## $ visits <dbl> 13, 19, 18, 18, 12, 19, 42, 19, 25, 24, 34, 28
```

Wide Data

As a comparison, let's also load a wide version of this dataset. {.codesmall}

```
wide_er <- read_csv(file =  
  "https://jhudatascience.org/intro\_to\_r/data/CO\_heat\_er\_visits\_DenverBoulder\_  
  
## Rows: 2 Columns: 13  
## — Column specification —————  
## Delimiter: ","  
## chr  (1): county  
## dbl (12): 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, .  
##  
## □ Use `spec()` to retrieve the full column specification for this data.  
## □ Specify the column types or set `show_col_types = FALSE` to quiet this mess
```

Wide vs Long Data: Which is better for plotting?

```
head(long_er)
```

```
## # A tibble: 6 × 3
##   county  year visits
##   <chr>   <dbl>   <dbl>
## 1 Boulder 2012     13
## 2 Boulder 2014     19
## 3 Boulder 2016     18
## 4 Boulder 2018     18
## 5 Boulder 2020     12
## 6 Boulder 2022     19
```

```
head(wide_er)
```

```
## # A tibble: 2 × 13
##   county `2011` `2012` `2013` `2014` `2015` `2016` `2017` `2018` `2019` `2020`
##   <chr>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Boulder 4.03  4.08  3.79  6.29  4.76  5.68  3.51  5.07  3.71  3.51
## 2 Denver  7.11  6.79  2.95  3.56  3.84  6.18  3.32  5.81  4.54  4.54
## #   2 more variables: `2021` <dbl>, `2022` <dbl>
```

Make a plot of visit rates by year for different counties

```
esquisser(wide_er) # county as x...? Tricky!  
esquisser(long_er) #county as x, visit rate as y, year as fill
```

GUT CHECK!

Why use Esquisse?

- A. Explore your data
- B. Get a “head start” on your code
- C. Both of these!

Some Alternatives to **esquisse**

- ggquicked: <https://smouksassi.github.io/ggquicked/>
- ggraptR: <https://github.com/cargomoose/ggraptR/>
- autoplot can be helpful for some packages (see [this blog post](#))

Summary

- Use Esquisse:
 - `library(esquisse)`
 - `esquisser()` function on a dataset
- Use the `viewer = "browser"` argument to launch in your browser.
- Code from Esquisse can be copied into code chunks to be generated in the "Plots" pane
- It's easier if your code is in "long" form!

Lab

- ▢ [Class Website](#)
- ▢ [Lab](#)
- ▢ [Day 6 Cheatsheet](#)



Image by [Gerd Altmann](#) from [Pixabay](#)